

3.1 *Theorem:*

$$\langle p | \hat{T} = \exp\left(\frac{-ibp}{\hbar}\right) \langle p |.$$

Proof. Given

$$\langle x | \hat{T} = \langle x - b |, \quad (1)$$

we can expand with

$$\begin{aligned} \langle p | \hat{T} &= \sum_x \langle p | x \rangle \langle x | \hat{T} \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \langle x - b | dx \end{aligned}$$

and if we let $\xi = x - b$ and subsequently $x = \xi + b$ and $dx = d\xi$,

$$\begin{aligned} \langle p | \hat{T} &= \int_{-\infty}^{\infty} \exp\left(\frac{-ip(\xi + b)}{\hbar}\right) \langle \xi | d\xi \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ip\xi}{\hbar}\right) \exp\left(\frac{-ipb}{\hbar}\right) \langle \xi | d\xi \\ &= \exp\left(\frac{-ipb}{\hbar}\right) \int_{-\infty}^{\infty} \exp\left(\frac{-ip\xi}{\hbar}\right) \langle \xi | d\xi \end{aligned}$$

and the second term is an equivalent definition of $\langle p |$ as ξ lies in position-space, simplifying to

$$\langle p | \hat{T} = \exp\left(\frac{-ipb}{\hbar}\right) \langle p |.$$

□

3.2 *Theorem:*

$$\langle p | \hat{D} = \frac{ip}{\hbar} \langle p |.$$

Proof. Given the definition

$$\langle x | \hat{D} = \frac{d}{dx} \langle x |, \quad (2)$$

we can once again expand

$$\begin{aligned} \langle p | \hat{D} &= \sum_x \langle p | x \rangle \langle x | \hat{D} \\ &= \int_{-\infty}^{\infty} \exp\left(\frac{-ipx}{\hbar}\right) \frac{d}{dx} \langle x | dx \end{aligned}$$

Using integration by parts, we can set

$$\begin{aligned} u &= \exp\left(-\frac{ipx}{\hbar}\right) & du &= -\frac{ip}{\hbar} \exp\left(-\frac{ipx}{\hbar}\right) dx \\ dv &= \frac{d}{dx} \langle x | dx & v &= \langle x | \end{aligned}$$

yielding the equation

$$\begin{aligned} \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \frac{d}{dx} \langle x | dx &= \exp\left(-\frac{ipx}{\hbar}\right) \langle x | \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\frac{ip}{\hbar} \exp\left(-\frac{ipx}{\hbar}\right) \langle x | dx \\ &= \exp\left(-\frac{ipx}{\hbar}\right) \langle x | \Big|_{-\infty}^{\infty} + \frac{ip}{\hbar} \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \langle x | dx \end{aligned}$$

As $\lim_{x \rightarrow \pm\infty} \exp(-ipx/\hbar) = 0$, both terms simplify to

$$\langle p | \hat{D} = \frac{ip}{\hbar} \langle p |.$$

□

3.3 The mirror reflection operator $\hat{R}$ produces the reflection about the origin of any function in x -space. It is called the parity operator in physics. Its x -space instruction is $\langle x | \hat{R} = \langle -x |$. What is its p -space instruction?

Suppose we have a state ψ ,

$$\begin{aligned} \langle p | \hat{R} | \psi \rangle &= \sum_x \langle p | x \rangle \langle x | \hat{R} | \psi \rangle \\ &= \int_{-\infty}^{\infty} \exp\left(-\frac{ipx}{\hbar}\right) \langle -x | \psi \rangle dx \end{aligned}$$

Now let $\xi = -x$, producing

$$\begin{aligned} \langle p | \hat{R} | \psi \rangle &= - \int_{-\infty}^{\infty} \exp\left(\frac{ip\xi}{\hbar}\right) \langle \xi | \psi \rangle d\xi \\ &= - \int_{\infty}^{-\infty} \exp\left(-\frac{i(-p)\xi}{\hbar}\right) \langle \xi | \psi \rangle d\xi \\ &= \langle -p | \psi \rangle \end{aligned}$$

3.4-1 Given $\hat{\tau} |p\rangle = |p + Q\rangle$ where $Q = \hbar/L$. *Theorem:*

$$\langle p | \hat{\tau} = \langle p - Q |.$$

Proof. $\langle p | \hat{\tau} = (\hat{\tau} |p\rangle)^* = \langle p | \hat{\tau}^\dagger = \langle p - Q |$ □

3.4-2 *Theorem:*

$$\langle x | \hat{\tau} = \exp\left(\frac{iQx}{\hbar}\right) \langle x |$$

Proof.

$$\begin{aligned}\langle x | \hat{\tau} &= \sum_p \langle x | p \rangle \langle p | \hat{\tau} = \sum_p (\langle p | x \rangle)^* \langle p | \hat{\tau} \\ &= \sum_p \exp\left(\frac{ipx}{\hbar}\right) \langle p - Q | \end{aligned}$$

Now let $p' = p - Q$,

$$\begin{aligned}\langle x | \hat{\tau} &= \sum_p \exp\left(\frac{i(p' + Q)x}{\hbar}\right) \langle p' | \\ &= \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \exp\left(\frac{iQx}{\hbar}\right) \langle p' | \\ &= \exp\left(\frac{iQx}{\hbar}\right) \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \langle p' | \\ &= \exp\left(\frac{iQx}{\hbar}\right) \langle x | \end{aligned}$$

□

3.5-1 *Theorem:*

$$\langle x | \hat{\tau} \hat{D} | \psi \rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{d\psi}{dx}$$

Proof. $\langle x | \hat{\tau} \hat{D} | \psi \rangle = \frac{d}{dx} \langle x | \hat{\tau} | \psi \rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{d}{dx} \langle x | \psi \rangle \quad \square$